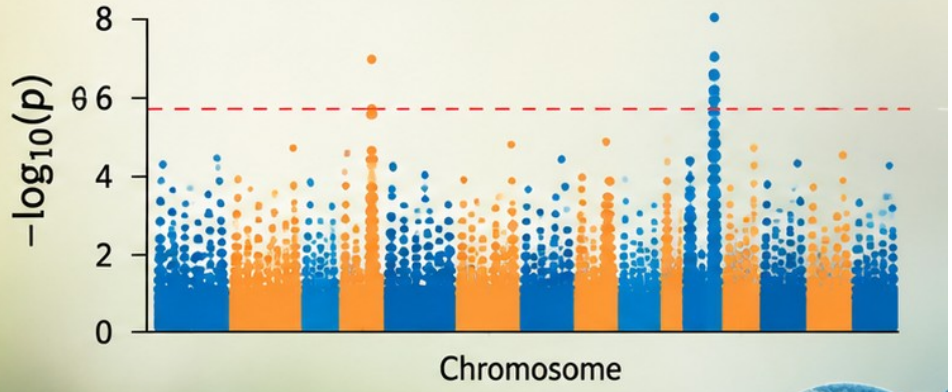
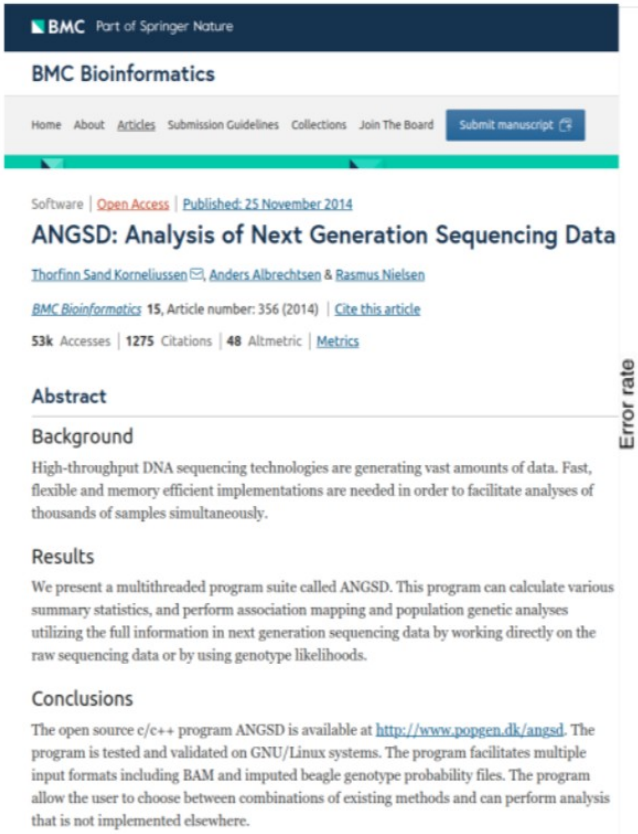
Genome-Wide Association Studies (GWAS) for Evolutionary Biology

Nikolay Oskolkov, Group Leader (PI) at NIRI, Riga, Latvia





ANGSD: genotype likelihood tool for low-coverage data



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ANGSD: Analysis of Next Generation Sequencing Data

Thorfinn Sand Korneliussen, Anders Albrechtsen & Rasmus Nielsen

BMC Bioinformatics 15, Article number: 356 (2014) | Cite this article

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Abstract

Background

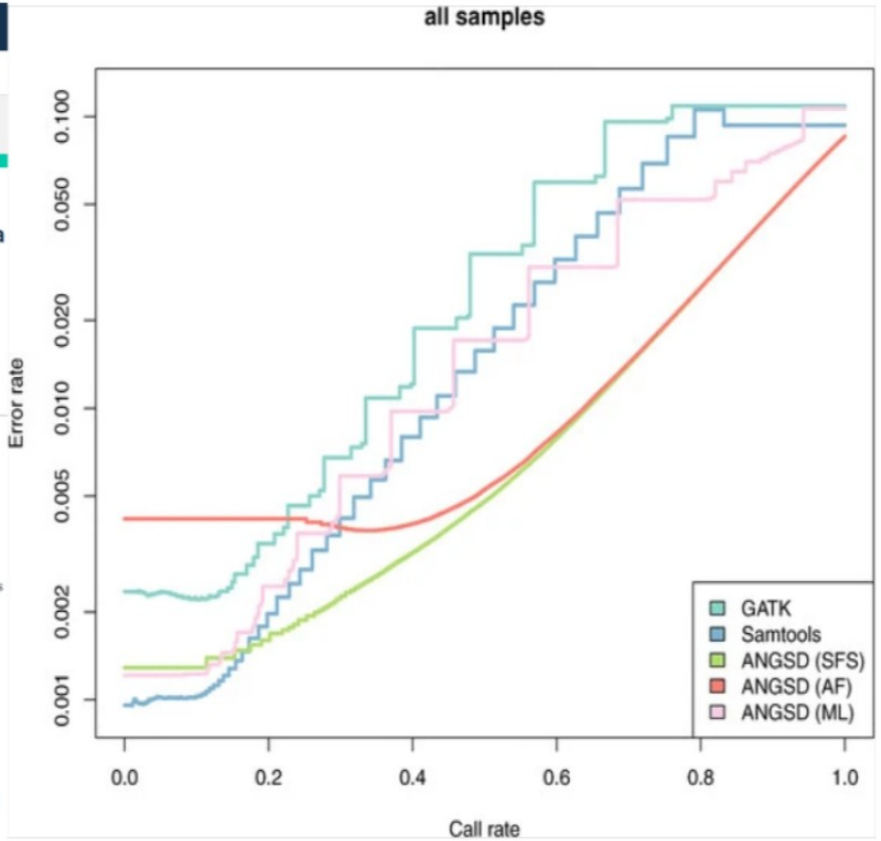
High-throughput DNA sequencing technologies are generating vast amounts of data. Fast, flexible and memory efficient implementations are needed in order to facilitate analyses of thousands of samples simultaneously.

Results

We present a multithreaded program suite called ANGSD. This program can calculate various summary statistics, and perform association mapping and population genetic analyses utilizing the full information in next generation sequencing data by working directly on the raw sequencing data or by using genotype likelihoods.

Conclusions

The open source c/c++ program ANGSD is available at http://www.popgen.dk/angsd. The program is tested and validated on GNU/Linux systems. The program facilitates multiple input formats including BAM and imputed beagle genotype probability files. The program allow the user to choose between combinations of existing methods and can perform analysis that is not implemented elsewhere.



(Korneliussen T, 2014)

Example of PCA plot from 1000G project

We will use ANGSD (Korneliussen T, 2014) and compute genotype likelihoods on 435 bam-files from the 1000G project.

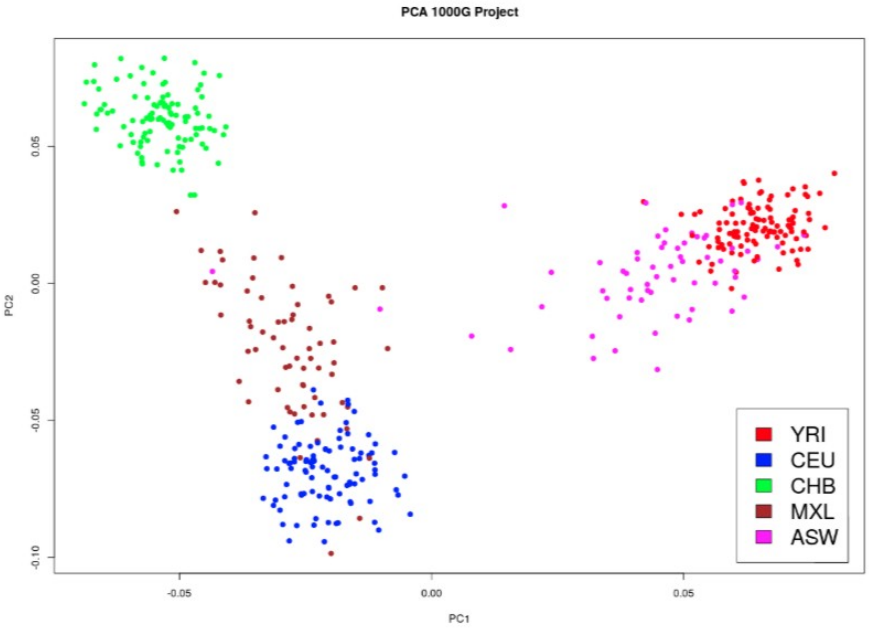
```
cd /home/nikolay/Documents/Teaching/PopGen_2023/data/1000G_bam_files
find $PWD -name '*.bam' > 1000G_bam_list.txt
angsd -bam 1000G_bam_list.txt -GL 2 -doMajorMinor 1 -doMaf 1 -SNP_pval 2e-6 \
-minMapQ 30 -minQ 20 -minInd 25 -minMaf 0.05 -doGlf 2 -out 1000G -P 5
```

Next, we will run PCAANGSD to infer admixures proportions:

```
pcangsd -b 1000G.beagle.gz -o pca_1000G -t 4
```

and finally we will plots the results using custom R scripts:

```
C <- as.matrix(read.table("pca_1000G.cov"))
e <- eigen(C)
pops <- readLines("1000G_bam_list.txt")
pops <- sapply(strsplit(pops, "\\."), function(x) x[6])
mycolor <- rep("red", length(pops))
mycolor[pops == "CEU"] <- "blue"
mycolor[pops == "CHB"] <- "green"
mycolor[pops == "MXL"] <- "brown"
mycolor[pops == "ASW"] <- "magenta"
plot(e$vectors[, 1:2], xlab = "PC1", ylab = "PC2", main = "PCA 1000G Project",
    col = mycolor, pch = 19)
legend("topright", c("YRI", "CEU", "CHB", "MXL", "ASW"), fill = c("red", "blue",
    "green", "brown", "magenta"), cex = 2, inset = 0.02)
```



Hard genotype calls vs. genotype likelihoods

Genotypes:	CC	CT	CC	CT	TT
Hard calls:	0	1	0	1	2
Likelihoods:	0.98 0.02 0.00	0.20 0.75 0.05	0.89 0.10 0.01	0.10 0.80 0.10	0.15 0.20 0.65

- Minor allele frequency based on hard calls:

$$\mathrm{MAF}_{\mathrm{hc}} = \frac{1+1+2}{2*5} = \frac{4}{10}$$

- Minor allele frequency based on genotype likelihoods:

$$\mathrm{MAF}_{\mathrm{gl}} = \frac{0*(0.98+0.20+0.89+0.10+0.15)+1*(0.02+0.75+0.10+0.80+0.20)+2*(0.00+0.05+0.01+0.10+0.65)}{2*5} = \frac{3.49}{10}$$

NGSadmix vs. ADMIXTURE

JOURNAL ARTICLE

Estimating Individual Admixture Proportions from Next Generation Sequencing Data

Line Skotte ✉, Thorfinn Sand Korneliussen, Anders Albrechtsen Author Notes

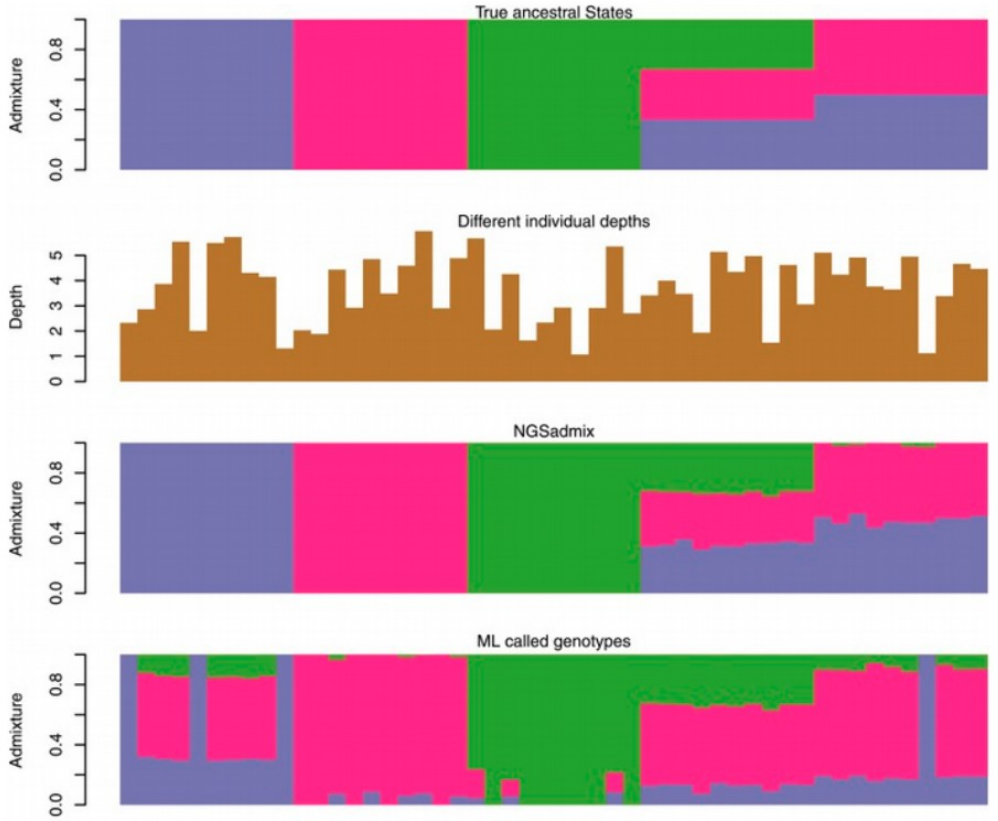
Genetics, Volume 195, Issue 3, 1 November 2013, Pages 693–702, https://doi.org/10.1534/genetics.113.154138

Published: 01 November 2013 **Article history** ▾

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Abstract

Inference of population structure and individual ancestry is important both for population genetics and for association studies. With next generation sequencing technologies it is possible to obtain genetic data for all accessible genetic variations in the genome. Existing methods for admixture analysis rely on known genotypes. However, individual genotypes cannot be inferred from low-depth sequencing data without introducing errors. This article presents a new method for inferring an individual's ancestry that takes the uncertainty introduced in next generation sequencing data into account. This is achieved by working directly with genotype likelihoods that contain all relevant information of the unobserved genotypes. Using simulations as well as publicly available sequencing data, we demonstrate that the presented method has great accuracy even for very low-depth data. At the same time, we



Example of admixture plot from 1000G project

We will use ANGSD (Korneliussen T, 2014) and compute genotype likelihoods on 435 bam-files from the 1000G project.

```
cd /home/nikolay/Documents/Teaching/PopGen_2023/data/1000G_bam_files
find $PWD -name '*.bam' > 1000G_bam_list.txt
angsd -bam 1000G_bam_list.txt -GL 2 -doMajorMinor 1 -doMaf 1 -SNP_pval 2e-6 \
-minMapQ 30 -minQ 20 -minInd 25 -minMaf 0.05 -doGlf 2 -out 1000G -P 5
```

Next, we will run NGSadmix to infer admixures proportions:

```
NGSadmix -likes 1000G.beagle.gz -K 3 -minMaf 0.05 -seed 1 -o 1000G
```

and finally we will plots the results using custom R scripts:

```
pops <- readLines("1000G_bam_list.txt")
pops <- sapply(strsplit(pops, "\\."), function(x) x[6])
source("https://raw.githubusercontent.com/GenisGE/evalAdmix/master/visFuns.R")
qopts <- read.table("1000G.qopt")
ord <- orderInds(pop = pops, q = qopts, popord = c("YRI", "ASW", "CEU", "MXL",
    "CHB"))
barplot(t(qopts)[, ord], col = c(3, 2, 4), las = 2, space = 0, border = NA)
text(sort(tapply(1:length(pops), pops[ord], mean)), -0.05, unique(pops[ord]))
abline(v = cumsum(sapply(unique(pops[ord]), function(x) {
    sum(pops[ord] == x)
})), col = 1, lwd = 1.2)
```

